THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 5–13: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity, Our Environment** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** nephron	Filtration unit
2	**(a)** Student A is correct	Turgidity of guard cells
3	**(d)** muscle cells during lack of oxygen	Causes cramps
4	**(c)** an electrical impulse	Nerve impulse
5	**(d)** cytokinin	High in fruits and seeds
6	**(a)** Correct	Reflex arc
7	**(c)** Hydra	Also yeast
8	**(b)** placenta	Villi
9	**(b)** DNA	Hereditary material
10	**(b)** 50%	X or Y sperm
11	**(c)** grass, goat and human	Producer → consumer
12	**(c)** 1%	Converted to food

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	False — the right side receives and pumps deoxygenated blood to the lungs.	2
14	Gap between nerve endings of two neurons; chemicals released at the axon end diffuse across and start an impulse in the next neuron.	2
15	False — self: pollen to stigma of the same flower; cross: pollen to stigma of another flower.	2
16	The transmission of characters (traits) from parents to offspring.	2
17	False — only 3–4 levels exist because energy decreases greatly at each level.	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Through xylem; root cells take in ions creating a pressure difference; transpiration pull — evaporation from leaves draws water up; also regulates temperature	3
19	Hyphae bear sporangia containing spores; spores are released and germinate on moist surfaces into new Rhizopus	3
20	F1: all violet (Vv); F2: VV, Vv, Vv, vv → 3 violet : 1 white	1+2
21	They make food from sunlight (autotrophs); all other organisms depend on them directly or indirectly	3
22	Round yellow × wrinkled green gives new combinations (round green, wrinkled yellow) in F2, showing the seed shape and colour genes are inherited independently	3
	OR: Mother: XX → all eggs X; father: XY → sperms X or Y; X sperm + X egg → girl (XX); Y sperm + X egg → boy (XY). 50% chance each.	3
23	Biodegradable: broken down by microbes (paper, peels); non-biodegradable: not broken down, persist (plastic, DDT); latter cause pollution	3
	OR: Energy available decreases at each level (90% lost as heat); after 3–4 levels little energy remains	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) cerebrum, mid-brain, pons, medulla, cerebellum, spinal cord, pituitary (ii) fore-brain: thinking, sensory/motor areas; mid-brain: reflex movements of head/eyes; hind-brain: cerebellum (balance), medulla (involuntary)	2+3
	OR: (i) central (brain, spinal cord) and peripheral (cranial and spinal nerves) (ii) reflex actions; conduction of impulses to/from brain (iii) communication between CNS and body parts	2+2+1

25	(i) only ~10% of energy passes to the next level (ii) producer 10 000 → primary 1000 → secondary 100 → tertiary 10 J (iii) less energy available	2+2+1
	OR: (i) layer of O_3 in the upper atmosphere (ii) CFCs used as refrigerants and in fire extinguishers (iii) UNEP agreement (1987) to freeze CFC production at 1986 levels (iv) skin cancer	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET B

Chapters 5–13: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity, Our Environment** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(b)** 4	Two atria, two ventricles
2	**(c)** ATP	Adenosine triphosphate
3	**(a)** thigmotropism	Touch
4	**(c)** vertebral column	Backbone
5	**(d)** testes	Male hormone
6	**(d)** urethra	Common passage
7	**(c)** fallopian tube (oviduct)	Zygote formed
8	**(b)** stamen	Anther and filament
9	**(b)** 1	rryy
10	**(a)** all tall	Tall is dominant
11	**(a)** Student A is correct	Refrigerants
12	**(b)** producers	First trophic level

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	The amount of dissolved oxygen in water is much lower than in air, so they breathe faster.	2
14	Auxin, gibberellin (cytokinin)	1+1
15	True — the lining becomes thick and spongy with blood supply each month.	2
16	A cross considering only one pair of contrasting characters (e.g. tall vs short).	2
17	A functional unit consisting of all living organisms (biotic) of an area interacting with the non-living (abiotic) components.	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Artificial kidney: blood passes through tubes in a dialysing fluid; nitrogenous wastes diffuse out; used when kidneys fail.	3
19	Through hormones secreted by endocrine glands directly into blood; e.g. adrenaline raises heart rate; insulin lowers blood sugar	1+2
20	Helps in survival under changing environments; basis for evolution; adaptation	3
21	Use of disposable items increases non-biodegradable waste; carry cloth bags; segregate waste; compost kitchen waste	1+2
22	Food chain is disturbed: organisms of the next level starve; the previous level grows unchecked; ecological imbalance	3
	OR: Segregation of biodegradable and non-biodegradable waste; composting; reduce, reuse and recycle; avoid plastic bags	3
23	O_2/CO_2 by diffusion via stomata; excess water by transpiration; wastes stored in leaves that fall, in vacuoles, as resins/gums in old xylem	3
	OR: Blood passes twice through the heart in one cycle — pulmonary (heart–lungs–heart) and systemic (heart–body–heart); keeps oxygenated and deoxygenated blood separate for efficient oxygen supply to warm-blooded animals	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) autotrophic — photosynthesis using sunlight, CO_2, water, chlorophyll (ii) CO_2 enters, O_2 leaves; water vapour lost (iii) stomata open at night, CO_2 stored as acid and used during day	2+2+1

	OR: (i) nostrils → pharynx → larynx → trachea (with cartilage rings) → bronchi → bronchioles → alveoli; diaphragm (ii) alveoli give a huge surface (~80 m²), thin walls, rich blood capillaries	3+2
25	(i) testes, scrotum, vas deferens, seminal vesicles, prostate, urethra, penis (ii) sperm formation requires a lower temperature (iii) carries sperms from testis to urethra	3+1+1
	OR: (i) fission (Amoeba), budding (Hydra), regeneration (Planaria) / spore formation (Rhizopus) (ii) single parent, no gametes, rapid multiplication	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET C

Chapters 5–13: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity, Our Environment** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(b)** sphincter muscle	Pyloric sphincter
2	**(d)** stomata	Guard cells
3	**(b)** medulla	Hind-brain
4	**(d)** synapse	Chemical transmission
5	**(b)** Student A is correct	Hibiscus is bisexual
6	**(b)** zygote	Fertilised egg
7	**(b)** 23	Half
8	**(c)** round yellow	Both dominant
9	**(d)** the temperature at which eggs are incubated	Environmental
10	**(c)** 75%	3/4
11	**(c)** producers	Autotrophs
12	**(a)** three oxygen atoms	O_3

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Ventricles pump blood to various organs (and lungs) with greater force.	2
14	False — it is due to rapid change in water content of cells, not growth.	2
15	It helps species survive changing environments; some variants may survive drastic changes.	2
16	False — females have a perfect pair (XX); males have a mismatched pair (XY).	2
17	The various steps in a food chain at which transfer of food/energy takes place.	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Transport of soluble food via phloem; supplies food to all parts including storage organs, fruits; uses ATP energy to load sucrose, osmotic pressure moves it	1+1+1
19	It receives information from all receptors, processes it and sends instructions; fore-brain, mid-brain, hind-brain	2+1
20	Maintains body design; copying reactions are not absolutely accurate, giving slight variations useful for evolution	2+1
21	Less waste, saves resources and energy, more hygienic if cleaned properly (kulhads vs plastic cups), cheaper in the long run	3
22	Bowman's capsule, glomerulus, tubule, collecting duct; filtration in glomerulus, selective reabsorption of glucose, amino acids, salts, water in tubule → urine	1½+1½
	OR: Acid: activates pepsin, kills bacteria; liver: secretes bile which neutralises acid and emulsifies fats	1½+1½
23	Secreted directly into blood; heart beats faster supplying more O_2; blood diverted to skeletal muscles; breathing rate increases; body prepared to fight or flee	3
	OR: Nervous: electrical impulses, fast, short-lived, via nerves; hormonal: chemicals, slow, long-lasting, via blood	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) phototropism (ii) auxin (iii) auxin diffuses to the shaded side, cells there elongate faster, shoot bends towards light (iv) the shoot bends again towards light	1+1+2+1
	OR: (i) pituitary — growth hormone (dwarfism) (ii) diabetes (iii) insulin — pancreas (iv) insulin injections and controlled diet	2+1+1+1
25	(i) father's sperm (ii) XX × XY → XX (girl), XY (boy) (iii) 50% (iv) sex is a matter of chance; the mother is not responsible	1+2+1+1

	OR: (i) TT × tt → F1 Tt (all tall); F1 self-crossed: TT, Tt, Tt, tt → 3 tall : 1 short (ii) in a hybrid, only the dominant trait is expressed (iii) first and second filial generations	3+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND